

C5-1

$$1. (a) \frac{1}{\sqrt{2}} (|0_1, 1_2\rangle - |0_1, 0_2\rangle) \\ = \frac{1}{\sqrt{2}} |0_1\rangle (|1_2\rangle - |0_2\rangle)$$

可分解, 故不是纠缠态.

$$(b) \frac{1}{\sqrt{2}} (|0_1, 1_2\rangle - |1_1, 0_2\rangle)$$

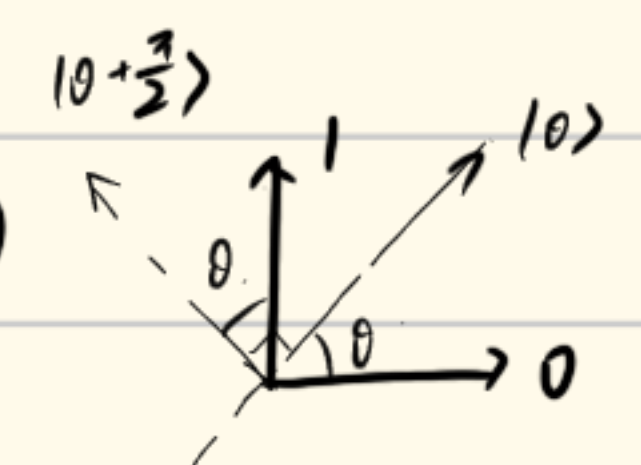
无法分解, 是纠缠态

$$(c) \frac{3}{5} |0_1, 1_2\rangle + \frac{4}{5} i |1_1, 1_2\rangle \\ = |1_2\rangle \left(\frac{3}{5} |0_1\rangle + \frac{4}{5} i |1_1\rangle \right)$$

可分解, 不是纠缠态.

$$(d) \frac{1}{2} (|0_1, 0_2\rangle + |0_1, 1_2\rangle + |1_1, 0_2\rangle + |1_1, 1_2\rangle) \\ = \frac{1}{2} [|0_1\rangle (|0_2\rangle + |1_2\rangle) + |1_1\rangle (|0_2\rangle + |1_2\rangle)] \\ = \frac{1}{2} (|0_1\rangle + |1_1\rangle) (|0_2\rangle + |1_2\rangle)$$

可分解, 不是纠缠态

2. (a) 
$$\begin{cases} |1\rangle = \sin\theta |0\rangle + \cos\theta |0+\frac{\pi}{2}\rangle \\ |0\rangle = \cos\theta |0\rangle - \sin\theta |0+\frac{\pi}{2}\rangle \end{cases}$$

$$\text{则 } |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_1, 0_2\rangle + |1_1, 1_2\rangle)$$

$$= \frac{1}{\sqrt{2}} [(\cos\theta |0_1\rangle - \sin\theta |0_1+\frac{\pi}{2}\rangle) (\cos\theta |0_2\rangle - \sin\theta |0_2+\frac{\pi}{2}\rangle) \\ + (\sin\theta |0_1\rangle + \cos\theta |0_1+\frac{\pi}{2}\rangle) (\sin\theta |0_2\rangle + \cos\theta |0_2+\frac{\pi}{2}\rangle)] \\ = \frac{1}{\sqrt{2}} [\cos^2\theta |0_1, 0_2\rangle - \sin\theta \cos\theta |0_1+\frac{\pi}{2}, 0_2\rangle - \sin\theta \cos\theta |0_1, 0_2+\frac{\pi}{2}\rangle \\ + \sin^2\theta |0_1+\frac{\pi}{2}, 0_2+\frac{\pi}{2}\rangle + \sin^2\theta |0_1, 0_2\rangle + \sin\theta \cos\theta |0_1+\frac{\pi}{2}, 0_2\rangle \\ + \sin\theta \cos\theta |0_1, 0_2+\frac{\pi}{2}\rangle + \cos^2\theta |0_1+\frac{\pi}{2}, 0_2+\frac{\pi}{2}\rangle] \\ = \frac{1}{\sqrt{2}} (|0_1, 0_2\rangle + |0_1+\frac{\pi}{2}, 0_2+\frac{\pi}{2}\rangle)$$

(b) 式中只有 $|0_1, 0_2\rangle$ 和 $|0_1+\frac{\pi}{2}, 0_2+\frac{\pi}{2}\rangle$ 两种态, 粒子1和2为正关联纠缠态, 则两个光子必定从偏振器同一侧出射.